

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

¹ • write F as $/$, G as $\backslash$

² • render $a \leq b$ as $\boxed{\frac{a}{b}}$

³ • **rewrite** $X \leq XF$ as $1 \leq F$

$$\begin{aligned} \boxed{\frac{X/}{Y}} &= \boxed{\frac{X}{Y\backslash}} && \text{adj} \\ \\ \boxed{\frac{X/}{X/}} &\xRightarrow{\text{adj}} \boxed{\frac{X}{X\wedge}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\wedge}}_{\text{unit}} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\wedge\backslash}} \\ &\xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\wedge/}} \\ \\ \boxed{\frac{Y\backslash}{Y\backslash}} &\xRightarrow{\text{adj}} \boxed{\frac{Y\vee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\vee}{*}}_{\text{counit}} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\vee/}{/}} \\ &\xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\vee\backslash}{\backslash}} \end{aligned}$$

now we have $\wedge/ = /$ and $\vee\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

	F → G	F 's type	monad FG	1 ⊆ FG	GF ⊆ 1	F preserves u	G preserves n	FGF=F	GFG=G
1	$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \subseteq \triangleleft \triangleright$	$\triangleright \triangleleft \subseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
2	$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \subseteq \multimap \multimap$	$\multimap \multimap \subseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
3	$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$	$(R \cup S)^{\circ} = S^{\circ} \cup R^{\circ}$	$(R \cap S)^{\circ} = R^{\circ} \cap S^{\circ}$	$R^{\circ \circ} = R$	$R^{\circ \circ} = R$
4	$\circ \triangleleft \dashv \triangleright \circ$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \circ) = 1$	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \circ \otimes 1) = 1$	$=$	$=$	$=$	$=$
5	$\triangleright \circ \dashv \triangleleft \circ$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) = 1$ $(\triangleright \circ \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \circ) = 1$	$=$	$=$	$=$	$=$
6	$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \subseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
7	$\cup \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \subseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
8	$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \subseteq R$				
9	$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \subseteq (RS)/S$	$(T/S)S \subseteq T$	$(S \cup T)R = (SR) \cup (TR)$	$(R \cap R') / S = (R / S) \cap (R' / S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
10	$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \subseteq S \setminus (SR)$	$S(S \setminus T) \subseteq T$	$R(S \cup T) = (RS) \cup (RT)$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
11	$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \subseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \subseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
12	$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \subseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \subseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
13	$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \subseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \subseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
14	$\cdot f \dashv \cdot f^{\circ}$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$(S \cup T)R = (SR) \cup (TR)$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^{\circ} = (Xg^{\circ}) \cap (Yg^{\circ})$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
15	$f^{\circ} \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \subseteq f f^{\circ}$	$f^{\circ} f \subseteq 1$	$R(S \cup T) = (RS) \cup (RT)$	$f(X \cap Y) = fX \cap fY$	$ff^{\circ}f = f$	$f^{\circ}ff^{\circ} = f^{\circ}$
16	$i \dashv E$	$\text{Map} \multimap \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow PA$	$\exists : PB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
17	$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, PB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{f \exists}{\exists} = f \exists$	$\frac{R}{\exists} \exists = R$
18	$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, PB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\exists}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f} \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xleftrightarrow{\cdot T \dashv / T} \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
\text{f a map} \\
\begin{array}{c} \xleftrightarrow{\text{associativity}} \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f} \boxed{\begin{array}{c} XT \\ fR \end{array}} \xleftrightarrow{\cdot T \dashv / T} \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
\xrightarrow{\text{indirect equality}} \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
\end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = (fR)/S$$

(1.2b)

	$\bullet T$	$T \bullet$	$\bullet g$	$g^\circ \bullet$	Tn	$^\circ$	$\text{key} \triangleleft$	Δ	
1	$\bullet S$	$R / (S_1 \ S_2)$ $= R / S_2 / S_1$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$R / (Sg) =$ $(Rg^\circ) / S$	$\text{Map}(f) \implies f$ $(R / S) =$ $(f \ R) / S$	$---$	$(Y/S)^\circ =$ $S^\circ \setminus (Y^\circ)$	$(RS)^\wedge =$ $R^\wedge (S \otimes 1)$	$(R \cap R') /$ $S = (R / S)$ $\cap (R' / S)$
2	$S \bullet$	$S \setminus (R / W)$ $= (S \setminus R) / W$	$(S_1 \ S_2) \setminus R$ $= S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) =$ $(S \setminus R)g^\circ$	$(g^\circ S) \setminus R =$ $S \setminus (gR)$	$---$	$(S \setminus Y)^\circ =$ Y° / S°	$((S \otimes 1)R)^\wedge =$ $SR^\wedge$	$S \setminus (R \cap$ $R') = (S \setminus$ $R) \cap (S \setminus$ $R')$
3	$\bullet f$	$\text{Map}(f) \implies$ $R / (f \ S) =$ $(R / S) f^\circ$	$T \setminus (Rf^\circ) =$ $(T \setminus R) f^\circ$	$(R \ S)^\circ = S^\circ$ R°			$(R \ S)^\circ = S^\circ$ R°		$\text{Simple}(g)$ $\implies (X \cap Y)$ $g^\circ = (X \ g^\circ)$ $\cap (Y \ g^\circ)$
4	$f^\circ \bullet$	$\text{Map}(f) \implies f$ $(R / S) =$ $(f \ R) / S$	$(Tf^\circ) \setminus R =$ $f(T \setminus R)$	$---$	$(R \ S)^\circ = S^\circ$ R°		$(R \ S)^\circ = S^\circ$ R°		$f(T_1 \cap T_2) =$ $fT_1 \cap fT_2$
5	Rn	$---$	$---$	$---$	$R \Rightarrow S \Rightarrow T =$ $R \cap S \Rightarrow T$	$(R \Rightarrow Y)^\circ =$ $R^\circ \Rightarrow (Y^\circ)$	$---$	$R \Rightarrow S \cap T =$ $(R \Rightarrow S) \cap$ $(R \Rightarrow T)$	
6	$^\circ$	$(Y/T)^\circ =$ $T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ =$ Y° / T°	$(R \ S)^\circ = S^\circ$ R°	$(R \ S)^\circ = S^\circ$ R°	$(T \Rightarrow Y)^\circ =$ $T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ =$ $(R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv)$ both bends: the ^\circ section's display	$(T_1 \cap T_2)^\circ =$ $T_1^\circ \cap T_2^\circ$
7	$\text{key} \triangleleft$	$(RT)^\wedge =$ $R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge =$ $TR^\wedge$	$---$	$---$	$---$	$(R^\wedge)^\circ =$ $(R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv)$ both bends: the ^\circ section's display	$---$	$---$
8	u	$(S \cup T) R =$ $(S \ R) \cup (T \ R)$	$R (S \cup T) =$ $(R \ S) \cup (R \ T)$	$(S \cup T) R =$ $(S \ R) \cup (T \ R)$	$R (S \cup T) =$ $(R \ S) \cup (R \ T)$	$R \cap (S \cup T)$ $= R \cap S \cup R$ $\cap T$	$(R \cup S)^\circ =$ $S^\circ \cup R^\circ$		
9	$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$R \cap \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$		

§1.3. Adjoint triples

Beyond $\mathbf{u} \dashv \mathbf{n}$, which §Section 1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} & \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ & f^\circ \dashv \cdot f \dashv f \backslash \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §Section 1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \dashv \cdot S \backslash$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \implies R / (f \ S) = (R / S) f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = \text{PA}$ they are the image triple:

(1.3b)

	operator	acts by	name
1	$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
2	$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
3	$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§Section 1.1's $\mathcal{D} \dashv \cdot \tau$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = P(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R / \tau = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot \tau \dashv 1 \cap \cdot / \tau$$

Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \cdot \dashv \cdot \dashv \cdot$ has no end, and §Section 1.1's other row of that kind, $\circ \dashv \triangleleft \dashv \triangleright \dashv \circ$, goes the same way.